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Pressure Transient Analysis in Geothermal Reservoirs Using Deep Learning Techniques

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Abstract

In the well tests in the geothermal reservoir, the non-isothermal behavior caused by the injection of cold water can affect the pressure response of the reservoir, which has been rarely considered in the existing analytical and numerical well testing models. In this work, we propose a well testing surrogate model to predict the pressure response of the geothermal reservoir during a non-isothermal injection fall-off test. First, a numerical well testing model of a geothermal reservoir with a horizontal well is built in MATLAB Reservoir Simulation Toolbox (MRST). Then, trained on the solutions of the numerical well testing model, two surrogate models based on long short term memory (LSTM) is developed. A multi-stage training scheme is employed to preserve the consistency by imposing the physics correlation between the pressure difference and pressure derivative. The surrogate models can accurately predict the pressure difference and derivative with mean absolute percentage errors (MAPE) of 5.90% and 5.51%, respectively. Moreover, the surrogate models can achieve a 2.90×10^5 speedup over the numerical well testing model, which contributes to efficient pressure response predictions and reservoir parameter evaluations.

Introduction

Well testing is an effective method to infer reservoir properties from the reservoir response to changed production or injection conditions. When the pressure response is measured, then well test analysis is synonymous with pressure transient analysis (PTA) (Horne, 1990). Conventional PTA based on the analytical models has been usually applied to the isothermal oil and gas reservoir. However, the injection fall-off test in a geothermal reservoir is a non-isothermal process, as the injection of cold water into the high-temperature reservoir will induce temperature variations in the reservoir. The cold-water front (cold bank) propagates and forms a low-temperature zone surrounding the injection wellbore, as shown in Figure 1, where the water viscosity increases and the mobility decreases. This non-isothermal behavior affects the injectivity and the subsequent reservoir response. However, existing analytical PTA models assume constant fluid properties, making them not suitable for non-isothermal PTA in geothermal reservoirs. Numerical models based on geothermal simulators like TOUGH2 (Pruess et al., 1999) have been developed for non-

isothermal PTA in geothermal reservoirs. Mclean et al (McLean et al., 2016; McLean & Zarrouk, 2017) developed a numerical non-isothermal falloff testing model in TOUGH2 and examined the effects of the skin zone properties and well properties on the pressure response. Akresh et al (2018) build a non-isothermal PTA model in KAPPA. The sensitivity analysis of the pressure response on various reservoir properties (e.g., permeability, thickness, etc.) and operational parameters (e.g., injection temperature and rate) are performed. Their results indicated that conventional PTA assuming isothermal condition can underestimate the reservoir permeability and flow capacity by 67%. Kaya et al (2019) developed a multi-phase, multi-component non-isothermal PTA model in TOUGH2 and studied the influence of the presence of CO₂ in the geothermal fluid on the constant rate drawdown test. Nonetheless, all the non-isothermal numerical models mentioned above are vertical well models. The vertical wells have limited heat extraction capacity due to relatively small reservoir thickness compared with the length and width. With the development of drilling technology, the horizontal wells gain more attention in geothermal extraction (Jiansheng et al., 2022). However, the numerical non-isothermal horizontal well testing models are not well established yet.

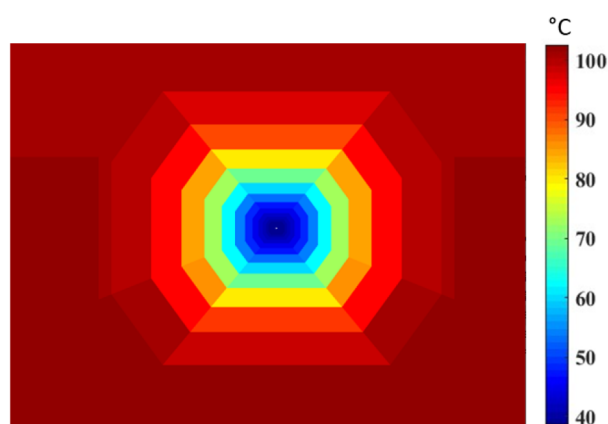


Figure 1—Cold bank effect.

In the traditional well testing interpretation, whether based on the analytical or numerical models, reservoir engineers need to manually change the parameters and match the pressure curves, which is quite time-consuming and may produce non-unique solutions. The automatic interpretation based on deep learning (DL) can significantly accelerate this procedure and mitigate the influence of human factors. In the literature, there are mainly two approaches of DL-based interpretations. The first approach is to directly train an inverse model where the pressure data is employed to predict the reservoir parameters. For example, Dong et al (2022) constructed an automatic interpretation model based on convolutional neural network (CNN). The input to the model is the dimensionless pressure derivative (p'_{wD}) while the outputs are the reservoir model parameters to be inversely estimated, including the dimensionless wellbore storage C_D , skin factor S , and permeability k . Other researchers have proposed similar workflows (Abdalla et al., 2024; Liu et al., 2020; Li et al., 2020). However, such workflow is based on the relationship between p'_{wD} and inversion parameters (C_D , S , k , etc.) given by the analytical PTA models in isothermal conditions. Therefore, it is not applicable to the interpretation of non-isothermal numerical well testing data.

The second automatic interpretation approach is to train a forward model to predict the pressure data and then construct an inverse model combining the forward model to estimate the reservoir parameters. Wang & Chen et al (2019) used long short term memory (LSTM) networks to predict the pressure response from the input of the gas flow rate in a tight reservoir. Chen et al (2022) developed a surrogate model based on bidirectional LSTM to forecast the pressure derivative in a fractured shale reservoir. But both studies only involve the forward model for pressure prediction without the inverse model to evaluate the reservoir parameters. In the subsequent work of Chen et al (2022), they constructed a bidirectional gated recurrent unit

(GRU) forward model to predict the pressure derivative and integrate it with deep reinforcement learning (DRL) based on twin delayed deep deterministic policy gradient (TD3) algorithm for the pressure data interpretation and reservoir parameter estimation. However, the forward model in their study is based on the semi-analytical well testing model with assuming constant fluid properties. Besides, they only predicted pressure derivative and used it as the only curve fitting objective in the inverse modelling. Nonetheless, both the pressure difference and pressure derivative data are used for the curve fitting and parameter estimation in the field application.

In this work, we propose a numerical model of non-isothermal PTA in a geothermal reservoir with a horizontal well. The water properties, especially the viscosity, is considered as a function of temperature. Further, we construct two surrogate models based on LSTM architecture to efficiently predict the pressure difference and derivative. Particularly, since the pressure difference and pressure derivative are mathematically correlated, we propose a multi-stage training scheme. These two surrogate models are firstly trained separately and then trained jointly with a physics constrained joint loss function, such that the correlation between the predicted pressure difference and pressure derivative is preserved. The surrogate models can be further employed to estimate the reservoir parameters in the future. Besides, we also develop a surrogate model based on fully connected neural network (FCNN) to predict the volume of the cold bank zone.

Methodology

Numerical Well Testing Model for Geothermal Reservoirs

To establish the numerical well testing model for geothermal reservoirs, the following physics assumptions are made:

1. The reservoir is homogeneous and isotropic, with a closed boundary and a constant boundary temperature.
2. The fluid is a slightly compressible single-phase fluid, and its flow follows Darcy's law.
3. The pressure loss along the wellbore is neglected, and the pressure is assumed to be uniform at all points within the wellbore.
4. The well is operated under a constant injection rate.
5. The wellbore storage effect is considered.

Mass conservation. The mass conservation equation for single-phase flow in geothermal reservoirs can be expressed as:

$$\frac{\partial \rho_f \phi}{\partial t} + \nabla \cdot (\rho_f \vec{v}) - q_f \rho_f = 0 \quad (1)$$

where ρ_f is the density of the fluid; ϕ is the porosity; t is the time; $\vec{v}$ is the Darcy velocities of the fluid; q_f is the volumetric sink and source terms.

The expression of Darcy velocities in Eq. (1) can be written as:

$$\vec{v} = -\frac{k}{\mu_f} (\nabla p - \rho_f g \nabla z) \quad (2)$$

where k is the permeability; μ_f is the viscosity of the fluid; p is the pressure; g is the gravitational acceleration; z is the height.

The well equation can be written as:

$$q = \frac{WI}{\mu_f} (p - p_{wf}) + C \frac{\partial p_{wf}}{\partial t} \quad (3)$$

$$WI = \frac{2\pi Lk}{\ln(r_e/r_w) + S} \quad (4)$$

where WI is the well index; C is the wellbore storage coefficient; p_{wf} is the bottomhole pressure; L is the well length; r_e is the equivalent radius; r_w is the well radius; S is the skin factor.

Energy conservation. The energy conservation equation for single-phase flow in geothermal reservoirs can be expressed as:

$$\frac{\partial [\rho_f \phi U_f + \rho_r (1 - \phi) U_r]}{\partial t} + \nabla \cdot (\rho_f \vec{v} U_f) + \nabla \cdot (p \vec{v}) - \nabla \cdot (\lambda_r \nabla T + \lambda_f \nabla T) = Q_f h_f \quad (5)$$

$$U_f = C_f T \quad (6)$$

where U_f and U_r are the fluid energy density and rock energy density, respectively; ρ_r is the density of the rock; λ_f and λ_r are the fluid thermal conductivity and rock thermal conductivity, respectively; T is the temperature; $Q_f h_f$ is the energy source and sink; C_f is the specific heat capacity of the fluid.

Then, the finite volume method is employed to discretize the mass and energy conservation equations, and Newton-Raphson iteration is used to solve the transient bottomhole pressure. During the iterative process, fluid (water) properties such as density and viscosity are updated according to the equation of state (EoS), accounting for the effects of pressure and temperature.

Surrogate Model for Pressure Response Prediction

Architecture. We develop two surrogate models based on LSTM to predict the time series of reservoir pressure response, including the pressure difference (Δp) and the pressure derivative ($\Delta p'$). Mathematically, this procedure can be expressed as,

$$\hat{y} = S(x; \theta) \quad (7)$$

where $\hat{y}$ is the predicted pressure difference (Δp) or derivative ($\Delta p'$) and S is the corresponding surrogate model; x is the input feature vector to the model; θ is the learnable parameter in LSTM, including bias and weight.

The structure of a LSTM memory block is shown in Figure 2.

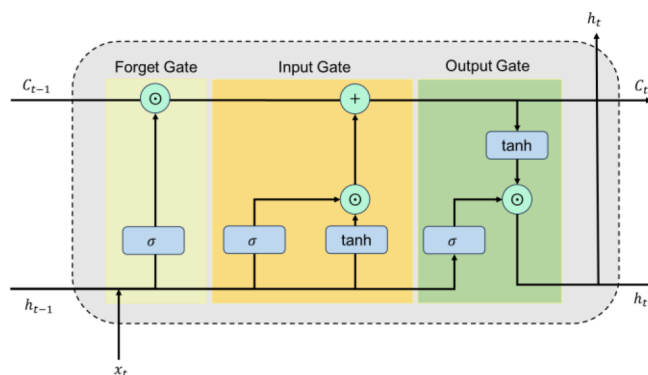


Figure 2—Structure of a LSTM memory block.

A LSTM memory block contains the forget gate, input gate and output gate. The forget gate determines which information should be discarded from the cell state. The output of this gate (f_t) ranges between 0 and 1, with 0 indicating complete forgotten and 1 representing complete retention of previous cell state information, which can be expressed as,

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (8)$$

where σ is the sigmoid function; W_f is the weight matrix of forget gate; h_{t-1} is the hidden state of previous time step and t represent the current time step; x_t is the input at current time step; b_f is the bias vector of forget gate.

The input gate controls the incorporation of new information through two components: i_t that determines which values to update and $\bar{C}_t$ that creates candidate values for potential addition to the cell state. The two components can be formulated as,

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (9)$$

$$\bar{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (10)$$

where $\tanh$ is the hyperbolic tangent function.

The cell state is subsequently updated as,

$$C_t = f_t \odot C_{t-1} + i_t \odot \bar{C}_t \quad (11)$$

where $\odot$ is the element-wise multiplication operation.

The procedure enables the selective preservation of long-term dependencies while incorporating relevant new information.

Finally, the output gate (o_t) regulates which components of the cell state should influence the hidden state output, expressed as,

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (12)$$

The final hidden state h_t can be computed as,

$$h_t = o_t \odot \tanh(C_t) \quad (13)$$

Training Procedure. The pressure derivative can be calculated from pressure difference and time, given by Bourdet et al (Bourdet et al., 1983) as,

$$\Delta p' = F(\Delta p, \ln t) = \frac{d(\Delta p)}{d(\ln t)} \quad (14)$$

Therefore, to preserve the relationship between the predicted pressure difference and pressure derivative, we employ a two-stage surrogate model training scheme in this work. In the first stage, we independently train the surrogate models for Δp and $\Delta p'$ predictions, respectively. The mean square error (MSE) is used as the loss functions, defined as,

$$\mathcal{L} = \frac{1}{N_b} \sum_{j=1}^{N_b} \frac{1}{N_p} \sum_{i=1}^{N_p} (\hat{y}_i - y_i)^2 \quad (15)$$

where $\hat{y}$ and y are the prediction and ground truth of pressure difference or derivative; i represents the i th time step and N_p is the total number of time steps; N_b is the batch size.

In the second stage, we train the two surrogate models jointly with one joint loss function L_J , defined as,

$$\mathcal{L}_J = \mathcal{L}(\Delta p) + \mathcal{L}(\Delta p') + \lambda \cdot (\widehat{\Delta p'} - F(\widehat{\Delta p}, \ln t)) \quad (16)$$

where λ is the regularization coefficient.

The third term in L_J (Eq. 16) considers the mismatch between the pressure derivative predicted and the pressure derivative calculated from the predicted pressure difference and time, which can enforce the physics correlation between the predicted pressure difference and derivative. $F(\widehat{\Delta p}, \ln t)$ is implemented by the automatic differentiation in Pytorch (Paszke et al., 2019).

Surrogate Model for Cold Volume Prediction

A surrogate model (denoted as S_c) based on fully connected neural network (FCNN) is constructed to predict the cold volume (V) between the cold-water bank and the wellbore. An FCNN architecture consists of a series of densely connected layers. There are 8 hidden layers in the cold volume surrogate model with 30 neurons on each layer. Each neuron in a hidden layer receives the outputs from every neuron in the preceding layer as its inputs. These inputs are summed, often with a bias term, and the result is passed through a nonlinear activation function σ , which allows the network to learn the complex, nonlinear relationships in the data. This process can be formulated as,

$$\hat{V} = S_c(\mathbf{x}; \theta) = \sigma(L_n \sigma(\dots \sigma(L_2 \sigma(L_1(x))) \dots)) \quad (17)$$

where $\hat{V}$ is the predicted cold volume; L_i is the i th hidden layer.

Numerical Experiment

To generate the dataset for the surrogate model training, we build a non-isothermal geothermal injection-falloff well testing model in MATLAB Reservoir Simulation Toolbox (MRST) (Lie & Møyner, 2021). The reservoir is homogeneous with a size of $10000 \times 10000 \times 100 \text{ m}^3$ in x , y and z directions. A horizontal injection well is located in the center of the reservoir along the x -axis direction. Radial grid and Cartesian grid are used to discretize the reservoir, as shown in Figure 3. The surface temperature is 30°C , and the reservoir temperature is calculated based on reservoir depth, geothermal gradient, and the surface temperature.

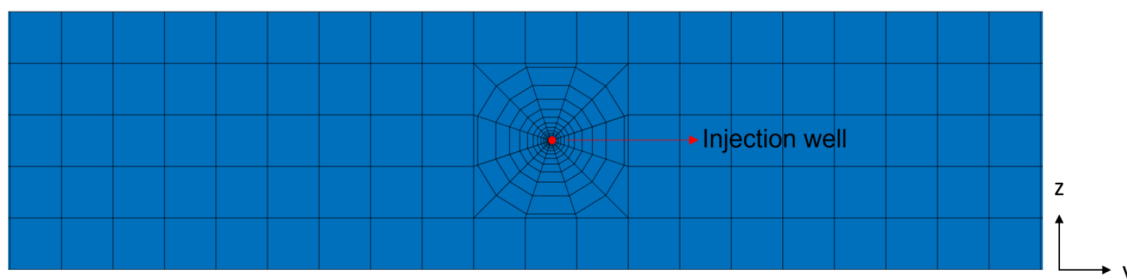


Figure 3—Reservoir discretization in the y - z plane (horizontal well along the direction into the y - z plane).

We select 13 reservoir model parameters as input to the surrogate model, including reservoir thickness (h), permeability (k), porosity (ϕ), thermal conductivity (λ), specific heat capacity (C_p), initial reservoir pressure (p_i), initial reservoir temperature (T_i), well length (L_w), wellbore storage coefficient (C), injection temperature (T_{inj}), injection rate (q_{inj}), injection duration (t_{inj}) and the time sequence after well shut-in (t). Noted that the time sequence is input to the surrogate model only to enable the automatic differentiation for pressure derivative calculation. We sample the first 12 input parameters using Latin Hypercube sampling and generate 1000 cases. Specifically, we don't directly sample p_i and T_i . Instead, we sample the pressure gradient (G_p), geothermal gradient (G_T) and reservoir depth (D) and then calculate p_i and T_i , which can ensure more realistic reservoir conditions. We also constrain that T_{inj} is lower than T_i . The range of each parameter is listed in Table 1. Further, we generate 1000 simulation cases and perform the numerical well-testing simulation based on MRST. After running all the cases in MRST, we collect the cold bank volume, the pressure difference, and pressure derivative data to train and test the corresponding surrogate models.

Table 1—Ranges of parameters for sampling.

Parameters	Lower bound	Upper bound
h (m)	15	50
k (mD)	20	1600
ϕ	0.02	2
λ (W/m/K)	1.6	3
C_p (J/kg/K)	700	1000
G_p (MPa/km)	9	12
G_T (°C/km)	30	60
D (km)	0.5	3
L_w (m)	150	1000
C (m ³ /MPa)	0.001	1
T_{inj} (°C)	15	60
q_{inj} (m ³ /day)	200	1000
t_{inj} (year)	0.5	5

The dataset is split into training, validation and testing sets with proportions of 80%, 10% and 10%. Min-Max normalization is used to scale the inputs and output. Specifically, since the pressure data and cold bank volume data both have relatively large differences in magnitudes (10^{-5} - 10^{-2} and 1 - 10^6), we apply $\log_2$ transformation to the pressure data and $\log_{10}$ transformation to the cold bank volume data to narrow the ranges before Min-Max normalization and employ the transformed data in the surrogate model training.

The LSTM architecture of the surrogate models for Δp and $\Delta p'$ both have 2 hidden layers with a dimension of 128. The hyperparameters of the surrogate models, including the number of epochs, batch size, learning rate and decay rate in each training stage are listed in [Table 2](#).

Table 2—Hyper-parameters to train the surrogate models for Δp and $\Delta p'$.

Hyperparameters	Δp model training in stage 1	$\Delta p'$ model training in stage 1	Joint model training in stage 2
Epoch number	500	500	500
Batch size	6	8	16
Learning rate	2×10^{-3}	2×10^{-3}	4×10^{-5}
Decay rate	6×10^{-7}	5×10^{-7}	8×10^{-7}

As for the surrogate model of the cold bank volume, we set the batch size, the learning rate and the decay rate at 16, 0.01 and 0.0001, respectively. The total number of epochs is 1000, while the training will stop early where there is no improvement of the validation loss in 30 epochs.

Results

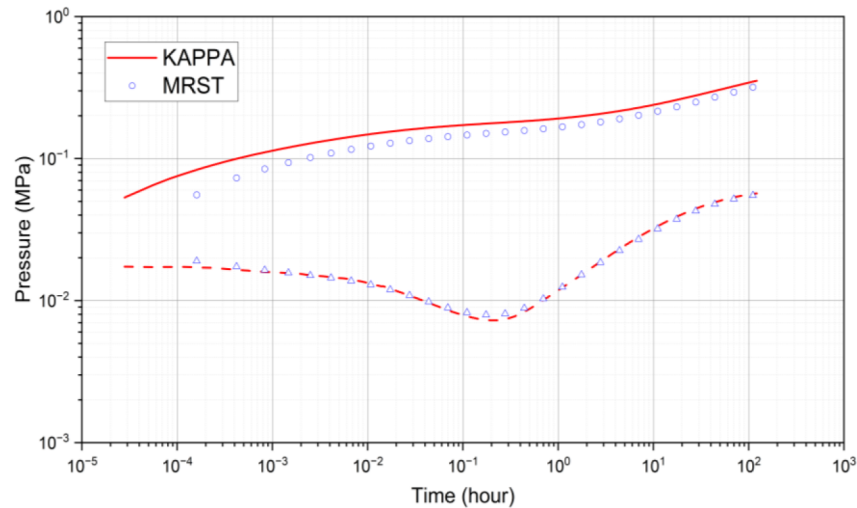
Well Testing Simulation Results

We firstly validate non-isothermal horizontal well testing model results in MRST with results in KAPPA. The setups for the validation model in MRST and KAPPA are given in [Table 3](#).

Table 3—Model setups.

Parameters	Values
Reservoir thickness h (m)	100.0
Permeability k (mD)	20.0
Porosity ϕ	0.2
Thermal conductivity of rock λ (W/m/K)	2.0
Specific heat capacity of rock C_p (J/kg/K)	800.0
Initial reservoir pressure p_i (MPa)	22.1
Initial reservoir temperature T_i (°C)	100.0
Well length L_w (m)	200
Wellbore storage coefficient C (m ³ /MPa)	0.01
Injection temperature T_{inj} (°C)	38.5
Injection rate q_{inj} (m ³ /day)	500.0
Injection time t_{inj} (year)	1.0

The pressure difference and derivative results of the numerical well-testing solution based on MRST and commercial solution based on KAPPA are exhibited in Figure 4. We can see a high agreement between the results of MRST and KAPPA, which guarantees the reliability of the training data obtained from MRST.

**Figure 4—Validation between MRST and KAPPA.**

Besides, we also compare the isothermal and non-isothermal well testing results. In the isothermal well testing simulation, we set up the injection temperature to be 100°C, which equals the reservoir temperature. The pressure responses are exhibited in Figure 5.

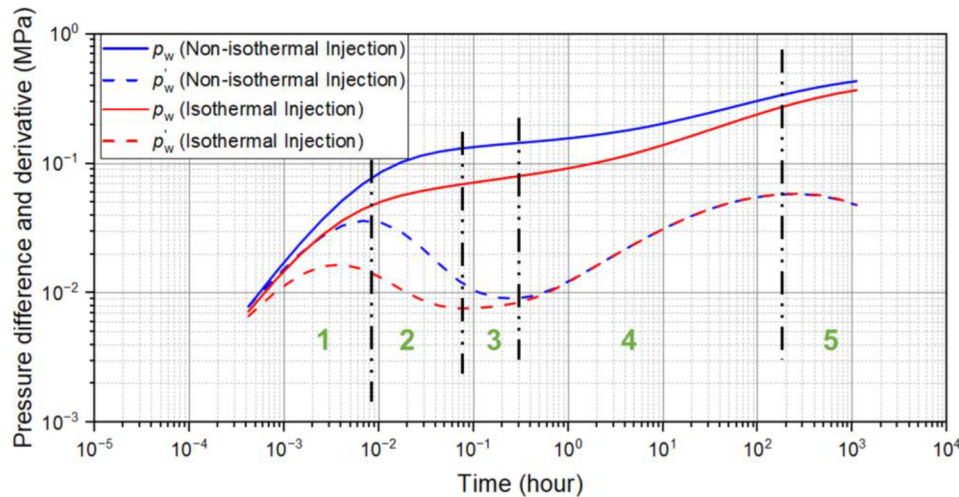


Figure 5—Comparison between isothermal and non-isothermal well testing.

From the pressure derivative curve of non-isothermal well testing model (blue dash curve), we can clearly identify different flow regimes as follows,

1. **Stage 1:** Wellbore storage effect. The pressure response is dominated by the wellbore storage effect in the early stage after the well shut in. Both the pressure and pressure derivative curves exhibit a straight line with a slope of 1.
2. **Stage 2:** Cold bank effect. The pressure derivative is expected to decrease significantly in this stage, as shown by the pressure derivative result of the isothermal well testing model (red dash line). Yet due to the presence of the low-temperature cold bank zone, the water mobility decreases, which leads to a rapid pressure decline and high pressure derivative values.
3. **Stage 3:** Transition. As the pressure front reaches the transition zone, where a thermal gradient exists between the cold bank and the original reservoir temperature zone, the change of pressure decline begins to slow down.
4. **Stage 4:** Upper and lower boundaries effect. Compared with the reservoir length and width, the reservoir thickness is relatively small. As a result, pressure propagates to the upper and lower boundaries of the reservoir, leading to a rapid pressure drop, which is reflected by a sharp increase in the pressure derivative.
5. **Stage 5:** Elliptical flow. After the pressure propagates to the upper and lower boundaries, it mainly spreads along the x- and y-directions. In the yz-plane, the pressure exhibits an elliptical propagation pattern. Because the reservoir length and width are relatively large, the rate of bottomhole pressure decline decreases, which is reflected by a reduction in the pressure derivative.

The isothermal and non-isothermal well-testing curves indicate that the influence of temperature on fluid properties significantly affects the transient pressure behavior. During the shut-in period after cold-water injection, both the pressure difference and pressure derivative under non-isothermal conditions are considerably larger than those under isothermal conditions. Therefore, when analyzing the pressure response characteristics in geothermal well testing, it is necessary to account for the non-isothermal behavior of water as well as the cold-bank effect.

Performance of Surrogate Model for Pressure Prediction

The non-dimensional coefficient of determination (R^2) is used as the metric to evaluate the model performance. The R^2 based on the testing dataset are 0.984 and 0.987 for the pressure difference and pressure derivative, respectively, as shown in Figure 6(a) and (b). The mean absolute percentage errors (MAPE) based

on all the testing cases are $(5.90 \pm 3.15)\%$ and $(5.51 \pm 2.70)\%$ for the pressure difference and derivative, respectively, as exhibited in Figure 6(c) and (d). The decent R^2 and MAPE results demonstrate the good prediction performance of both surrogate models and guarantee the high fidelity of using the surrogate models for pressure data encoding in the well tests.

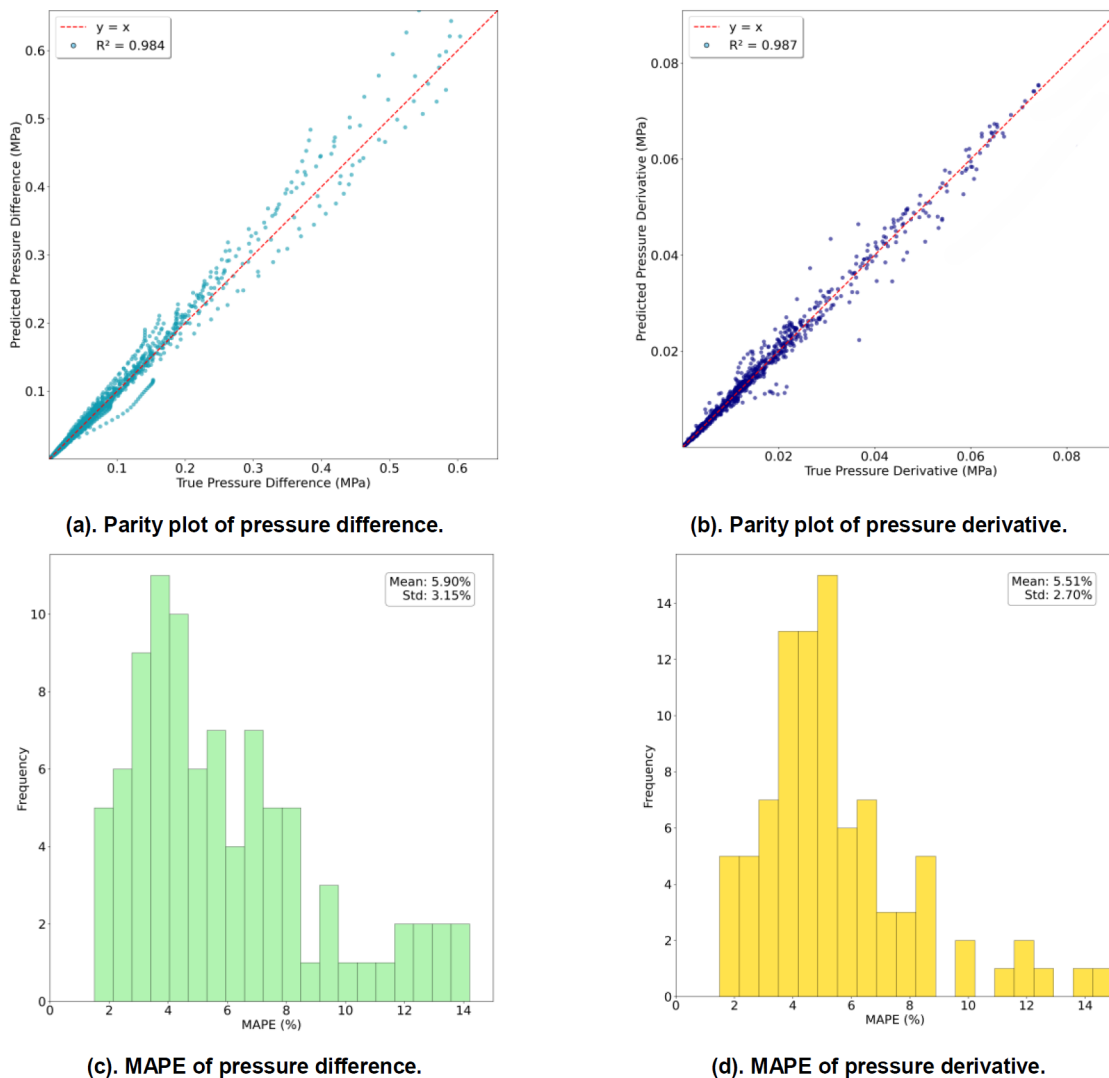


Figure 6—Performance of pressure surrogate model.

We also compare the prediction and simulation results of three different testing cases (88, 223, 595) as exhibited in Figure 7. We can see high agreements between simulation and prediction results. The surrogate models, especially the surrogate model of the pressure derivative, can accurately capture various curve trends. Besides, the prediction is highly efficient with an average CPU time of 1.02 seconds per prediction, which is about 7941.17 faster than the simulation model (around 135 minutes per simulation).

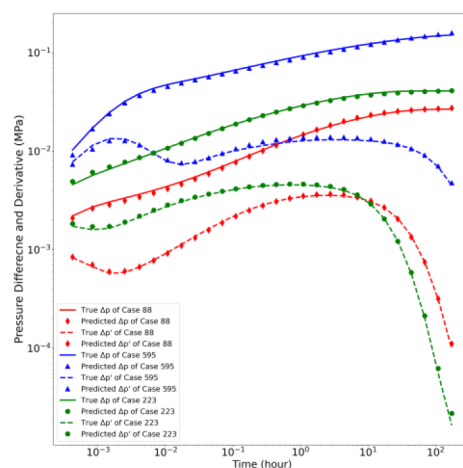


Figure 7—Matches between prediction and simulation results.

Performance of Surrogate Model for Cold Bank Volume Prediction

The surrogate model can predict the cold bank volume with decent accuracy. The R^2 is 0.967, as shown in Figure 8. The result also indicates a wide range of cold volume. For certain cases, the cold volume can reach $4 \times 10^5 \text{ m}^3$, which can significantly affect the pressure transient behavior. The cold bank volume surrogate model can help to roughly estimate the extent of cold water front propagation in the reservoir based on the predicted cold volume.

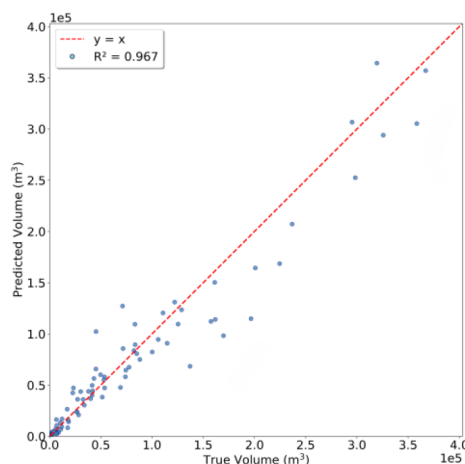


Figure 8—Performance of the surrogate model to predict the cold bank volume.

Conclusion

In this work, we develop a surrogate model for the non-isothermal injection-falloff pressure transient analysis (PTA) in a geothermal reservoir. The major conclusions are summarized below:

1. A numerical non-isothermal horizontal well testing model is built. The effect of the cold bank on the pressure response during the injection-falloff test in a geothermal reservoir is considered in the model, making it more suitable for the pressure transient analysis (PTA) in geothermal reservoirs with horizontal wells.
2. The well testing surrogate model based on the LSTM architecture can capture various trends of pressure derivative data, which can accurately predict the pressure difference and pressure derivative with mean absolute percentage errors (MAPE) of 5.90% and 5.51%, respectively. The surrogate models together also achieve a 7941.17 speedup over the numerical simulation model.

3. The surrogate model for the cold bank volume can perform prediction with a decent accuracy, in which the R^2 is 0.967. The cold bank volume predicted can help to roughly estimate the extent of cold water propagation in the reservoir.

Ultimately, the surrogate models can contribute to a more efficient PTA when evaluating the reservoir parameters, which will be carried out in our future work.

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